

NING WANG

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[Personal website](#) | [GitHub](#) | [Google Scholar](#)

Ph.D. researcher developing continual reinforcement learning methods for adaptive edge networks. Seeking Summer 2027 research and applied science internships in machine learning and network systems.

EDUCATION

University of Massachusetts Amherst

Ph.D., Electrical and Computer Engineering (in progress) | Feb 2024 - Present

Advisor: Prof. Beatriz Lorenzo

M.S., Electrical and Computer Engineering | Sep 2022 - Jan 2024 | GPA: 3.914/4.0

University of Electronic Science and Technology of China

B.S., Communication Engineering | Sep 2018 - Jul 2022 | GPA: 3.50/4.0

RESEARCH EXPERIENCE

Research Assistant | Network Science Lab, UMass Amherst

Department of Electrical and Computer Engineering | Sep 2023 - Present

Continual reinforcement learning for content reuse | IEEE TMC 2026 | [Code](#)

- Developed a continual reinforcement learning framework that uses a shared knowledge base and environment features to transfer learned policies to new network environments.
- Designed decisions for assigning users to servers, caching content, and reusing similar content instead of computing it again, subject to latency and resource constraints.
- Implemented Python/PyTorch training and evaluation with 5 drone-mounted servers, 15 users, and up to 10 environments. Published experiment scripts, saved results, figure-generation tools, and usage documentation.

Cooperative 360° video delivery | ImCDN

- Designed and implemented ImCDN, a simulation platform for cooperative tile-level 360° video delivery across edge nodes; evaluated content reuse and delivery performance under changing network conditions.
- First-author conference paper accepted to IEEE VTC-Fall 2026. Developing a journal extension on distributed lifelong learning for cooperative content delivery.

UAV mobility and task allocation | IEEE IoT Journal 2025 | [Code](#)

- Implemented a MATLAB framework combining particle swarm optimization for drone positioning, dynamic programming for task assignment, and lifelong learning for processing-resource allocation. Released training and online simulation modules, baseline comparisons, saved results, and paper-to-code documentation.

Satellite-terrestrial network control | IEEE TWC 2026

- Co-developed cross-domain lifelong learning for SDN configuration and resource allocation, sharing a knowledge base through domain-specific projections to balance age of information and energy consumption.
- Evaluated adaptation across heterogeneous learning domains; the published study reports up to 8 times faster convergence than baseline policy-gradient methods in complex domains.

PUBLICATIONS

Journal articles

N. Wang, Y. Wu, B. Lorenzo, S. Samarakoon, B. Liu. **Deep Lifelong Learning for Adaptive Semantic-Aware Content Reuse in UAV-Assisted Metaverse**. IEEE Transactions on Mobile Computing, vol. 25, no. 7, pp. 10847-10863, July 2026. [Paper](#)

Y. Wu, **N. Wang**, B. Lorenzo, S. Samarakoon, B. Liu. **Lifelong Learning-Based SDN Design for Dynamic Configuration and Resource Allocation in Satellite-Terrestrial Networks**. IEEE Transactions on Wireless Communications, vol. 25, pp. 21529-21544, 2026. [Paper](#)

N. Wang, Y. Wu, B. Lorenzo, B. Liu. **Semantic-Aware Architecture Design for a Lifelong Swarm Metaverse**. IEEE Internet of Things Journal, vol. 12, no. 9, pp. 12468-12482, May 2025. [Paper](#)

Conference papers

N. Wang, X. Liu, B. Lorenzo, S. Samarakoon. **ImCDN: An Immersive Content Delivery Network for Cooperative Tile-Level 360° Video Streaming**. IEEE VTC-Fall, Boston, MA, 2026. Accepted.

N. Wang, B. Lorenzo. **Semantic-Aware Blockchain Architecture Design for Edge-enabled Metaverse**. IEEE UEMCON, New York, NY, pp. 647-654, Oct. 2023. [Paper](#) | [Code](#)

Manuscript in preparation

N. Wang, B. Lorenzo, S. Samarakoon, B. Liu. ImCDN: Distributed Lifelong Learning for Cooperative 360° Video Content Delivery Networks. In preparation.

TECHNICAL SKILLS

Programming: Python, MATLAB, C, C++ | Machine learning: PyTorch | Research code: GitHub

Methods: Reinforcement learning, lifelong learning, dictionary learning, constrained optimization, network simulation

Technical writing: LaTeX | Languages: Chinese (native), English (proficient)

TEACHING EXPERIENCE

Teaching Assistant | University of Massachusetts Amherst

Sep 2023 - Present

Recent courses: ENGIN 112 - Introduction to Electrical and Computer Engineering; CS 250 - Introduction To Computation.

Other teaching: Quantum Computing; Security Engineering; Wireless Network Design; Continuous-Time Signals and Systems; Introduction to Artificial Intelligence.

- Lead ENGIN 112 discussion sections of 30-40 students, delivering five sessions every three weeks and guiding students through course concepts and questions.
- Independently design approximately one-third of course homework assignments and quizzes, collaborating with professors on course content and assessment.
- Previously led weekly CS 250 discussions for approximately 30 students and held one hour of office hours each week, providing individualized explanations and guidance.

PROFESSIONAL SERVICE

Session Chair, IEEE VTC-Fall 2026 | Session 1E: Authentication and Privacy | Sep 2026.

AWARDS

NSF Student Travel Grant and IEEE VTS Travel Grant, IEEE VTC-Fall 2026.